Xu **Han**

Software Engineer

Backend, Platform & AI



Profile

I work on the backend of an AI product in production: document processing, data and LLM agents. I focus on keeping it stable and its data correct. MSc from KTH. I measure before I conclude, and fix the mechanism rather than the symptom.

Contact

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Authorised to work in Switzerland (no visa sponsorship required)

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, C/C++

Backend: Node.js, NestJS, REST, SSE, WebSocket

AI & documents: LLM agents (Mastra, Claude API), MCP, Langfuse, OCR (Azure Document Intelligence)

Data: PostgreSQL (query plans, indexing), OpenSearch, S3

Infrastructure: Terraform, Docker, Linux, GitHub Actions

Observability: Prometheus, Grafana, Loki, OpenTelemetry

Security: OWASP Top 10, SSRF, least-privilege IAM, TLS

Frontend: React, Tailwind CSS

Languages

Chinese: native

English: C1

French: B1

Education & Certification

École 42 Paris
Software Engineering Programme — **May 2024 – Present**
Project-based, peer-reviewed curriculum: C/C++, Unix systems, networking, DevOps and web.

KTH Royal Institute of Technology, Stockholm
MSc in Engineering Physics — **Aug 2021 – Feb 2024**

Nankai University, Tianjin
BSc in Electronic Science and Technology — **Sep 2016 – Jun 2020**

Amazon Web Services
AWS Certified AI Practitioner — **Mar 2026**

Experience

KLAIRE, Paris *(LegalTech SaaS for notaries with an AI agent)*
Software Engineer (part-time) — **Oct 2026 – Present**
Software Engineer Intern — **Mar 2026 – Sep 2026**

- Built the AI assistant's document pipeline, turning case files of up to 1,800 documents (scans, PDFs, Word, Excel) into searchable text with OCR, and made it faster and more reliable, e.g. incremental imports, page-range OCR with retries, dedicated processing workers and an import query cut from 1.8 s to 63 ms.
- Built multi-tenant storage isolation: each law office has its own S3 bucket and an access key limited to its own files, so isolation is enforced by the cloud provider, not only by application code; Terraform provisions new offices automatically.
- Ran several data and infrastructure migrations, e.g. file storage from Azure Blob to S3 and then to one bucket per office, data and users onto the new production environment, the old chat product's users into the new app, and three overlapping file tables merged into one.
- Kept the backend stable: deploys that no longer interrupt users (each used to cause about 18 s of errors), monitoring and alerts (Prometheus, Grafana, Loki, OpenTelemetry), and production issues traced to their root cause and fixed.
- Debugged and hardened the AI assistant (LLM agents on Mastra and Claude) in production so that long tasks finish reliably, e.g. fixing runs that hung or looped, cutting memory leaks, saving progress after each step, and falling back to an EU-hosted model when the main one is overloaded.
- Built infrastructure and tooling, e.g. a self-hosted OpenSearch cluster for document search (Terraform, daily backups), an antivirus service that scans every public upload, and a backend test suite cut from about 40 to 2 minutes per shard.
- Worked on security hardening and fixes across the platform: access policies, environment isolation, HTTPS and patching.

Full-stack Software Engineer Intern *at inStorify* **Oct 2025 – Mar 2026**
Remote. Places pet-product brands in independent pet stores.

- Designed and built a centralised sales and billing system to replace monthly Excel calculations: one click generates and sends every store's Stripe invoice, a webhook keeps payment status in sync, and brand payouts are calculated by membership tier with a PDF statement each. Monthly processing went from about three days to ten minutes.

- Automated customer emails with AWS SES and scheduled jobs: pending-order reminders at 10, 13 and 20 days, out-of-stock alerts for admins, and reorder campaigns for stores inactive for three months.
- Also built the returns workflow, a quick-order flow with admin approval, a help desk with FAQ and support tickets, user analytics, and much of the dashboard redesign (Recharts, CSV export).
- Owned features end to end on Gadget.dev (Node.js) and React: data models, permissions, backend actions, UI and release to production.

Optical Engineer, MSc Thesis *at Tobii, Stockholm* **Jan 2023 – Jun 2023**

- Built Python automation and data-analysis pipelines (Pandas, Matplotlib) for polarisation-optics experiments on VR pancake lenses.

Research Assistant *at CAS* **Sep 2019 – Dec 2019**

Institute of Automation, Chinese Academy of Sciences

- Trained object-detection models (YOLO, SSD, Fast R-CNN), reaching 93.2% accuracy.

Projects

ft_transcendence *(full-stack web app, 42)*

github.com/IslandHanX/Transcendence

- Online Pong platform: a TypeScript API (Fastify, Prisma) with JWT sessions, Google sign-in and two-factor login by email code; live presence, game invitations and chat over WebSockets; tournaments, friends and match statistics, in English, French and Chinese.

Red Tetris *(multiplayer game, 42)*

github.com/IslandHanX/red-tetris-multiplayer

- Real-time multiplayer Tetris: a React client and a Node.js server (Express, Socket.IO) that gives every player the same piece sequence, sends penalty lines to opponents and hands over the host role when a player leaves; tested with Vitest under coverage thresholds.

Inception-of-Things *(Kubernetes, 42)*

- Built Kubernetes clusters with K3s on Vagrant VMs, routed three apps by host name through a Traefik ingress, and set up GitOps with Argo CD so that a push to Git redeploys the app automatically.